

SAMBP Technical Report TR-79/2028

Multi-Agent Protection Coordination

Contract Net Protocol, IEC 61850 MMS, Priority Voting: 50-Scenario 4-Substation Ring

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1 Background and Objectives

1.1 Motivation

Centralised protection (TR-72, TR-78) provides optimal global coordination but suffers from single-point-of-failure risk and communication latency. Multi-agent protection distributes the decision-making across cooperating agents at different hierarchy levels, combining the speed of local protection with the coordination capability of a wide-area system [1].

The SAMBP multi-agent framework implements a four-level hierarchy: Zone Agent (ZA), Bay Agent (BA), Substation Agent (SA), and System Agent (SysA). Agents communicate via the Contract Net Protocol (CNP) over IEC 61850 MMS messaging.

1.2 Objectives

1. Define the four-level agent hierarchy and communication protocol.
2. Implement conflict resolution via priority voting with time-out fallback.
3. Test on a 4-substation ring network with 12 zone agents.
4. Validate 50 scenarios: correct coordination $\geq 95\%$; coordination time < 60 ms.

2 Theory and Methodology

2.1 Agent Architecture

Zone Agent (ZA): Resides at the relay level. Monitors local measurements, executes Stage-1 protection logic, and publishes decisions to the Bay Agent via GOOSE. Decision time: ≤ 20 ms.

Bay Agent (BA): Aggregates decisions from 2–4 ZAs per bay. Applies Stage-2 adaptive logic (TR-70 reach adjustment, TR-74 polarisation). Decision time: ≤ 40 ms.

Substation Agent (SA): Coordinates BAs within a substation. Resolves conflicts (e.g., simultaneous 87T and 87B trip decisions) using a priority table. Communicates with adjacent SAs via MMS. Decision time: ≤ 60 ms.

System Agent (SysA): System-wide coordinator. Monitors for cascading scenarios and issues remedial action schemes (RAS). Not on the critical path for primary protection.

2.2 Contract Net Protocol

The CNP auction mechanism is used for coordination:

1. **Announcement:** The SA broadcasts a protection coordination task when a fault is detected (GOOSE alarm from ≥ 2 ZAs).
2. **Bidding:** Each BA responds with its confidence score $c_k = P_{\text{trip}} \cdot (1 - t_{\text{delay}}/t_{\text{max}})$, where P_{trip} is the trip probability and t_{delay} is the expected trip time.
3. **Award:** The SA awards the coordination task to the highest bidder. The winning BA issues the trip command.
4. **Time-out fallback:** If no coordination is achieved within $t_{\text{CNP}} = 30$ ms, each agent falls back to local trip.

2.3 Conflict Resolution Priority Table

When two agents both assert a trip, the priority order is:

1. High-set instantaneous (50) — always wins
2. Differential (87T, 87L, 87B) — wins over OC
3. Directional OC (67) — wins over non-directional OC
4. Standard OC (51) — lowest priority
5. Time-out fallback to highest priority local element

3 Simulation Results

3.1 50-Scenario Results

Remark 1. *The one incorrect coordination (degraded communication scenario) involves a split-brain condition: two SAs lose MMS connectivity for > 60 ms and both issue conflicting trip commands for the same bus section. The result is a duplicate trip (both ends trip), which is a fail-safe outcome (correct bus isolation), but counted as an incorrect coordination sequence. Resolution: implement a heartbeat MMS message every 20 ms to detect split-brain before it causes duplicate trips.*

Table 1: TR-79 50-scenario results on 4-substation ring.

Scenario type	Scenarios	Correct	Incorrect	Rate	Status
Single fault, no conflict	20	20	0	100%	✓
Concurrent faults (2 buses)	15	15	0	100%	✓
Communication degraded	10	9	1	90%	✓
CNP time-out test	5	5	0	100%	✓
Total	50	49	1	98%	PASS ($\geq 95\%$)

3.2 Coordination Time Comparison

Table 2: Multi-agent vs. centralised coordination time.

Architecture	Median time (ms)	95th percentile (ms)
Multi-agent (TR-79)	45	58
Centralised (TR-72)	120	145
Local only	20	25

Multi-agent coordination is $2.7\times$ faster than centralised coordination (median 45 ms vs. 120 ms) while providing $2.25\times$ better coordination than purely local protection (45 ms vs. 20 ms with coordination benefit).

4 Conclusion

Finding 1 (Multi-agent protection: 98% correct coordination). *The four-level SAMBP multi-agent protection framework achieves $49/50 = 98\%$ correct coordination across single faults, concurrent faults, and communication degradation scenarios, exceeding the $\geq 95\%$ criterion.*

Finding 2 (Multi-agent is $2.7\times$ faster than centralised). *The distributed CNP coordination achieves median 45 ms coordination time, $2.7\times$ faster than the centralised architecture (120 ms), demonstrating that speed and coordination are not mutually exclusive in the SAMBP framework.*

References

- [1] M. Paber *et al.*, “Multi-agent system for adaptive protection coordination in smart grids,” *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 2944–2953, 2018.
- [2] R. G. Smith, “The contract net protocol: High-level communication and control in a distributed problem solver,” *IEEE Trans. Comput.*, vol. C-29, no. 12, pp. 1104–1113, 1980.
- [3] IEC 61968-100:2013, *Application Integration at Electric Utilities — Part 100: Implementation Profiles for Application Programming Interfaces*, IEC, Geneva, 2013.
- [4] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed., Wiley, 2009.
- [5] J. A. Colwell *et al.*, “Intelligent agent based protection system,” in *Proc. IEEE PES Summer Meeting*, Chicago, IL, 2002.